Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/30/2026
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (80NSSC24K1274)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Worcester	**State:**	MA
Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	07/30/2024	**End Date:**	07/29/2027
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Hanson, Ian	**Contact Phone:**	
Contact Email:	ian.m.hanson@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 07/29/2027 per the PI and NSSC documentation (Ed., 3/30/26).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC24K1274		
Performance Goal No.:			
Performance Goal Text:			

Robert Hyers is U.S. Team Member on European Space Agencey (ESA)-sponsored projects, "Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC)," "Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS)," and "Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL)."
Funding is for Dr. Hyers's role as Team Member on the subprojects described below, associated with European Space Agency (ESA)-sponsored main projects as noted.

Main project: Peritectic Alloy Rapid Solidification with Electromagnetic Convection (PARSEC); Science Coordinator is Dr. Thomas Volkmann, German Aerospace Center/DLR, Institut für Materialphysik im Weltraum, Linder Höhe, Köln,

Task Description:	Germany. Subprojects: 1. Transformation Kinetics-ISS (International Space Station). Task description: Using results from thermophysical property evaluation and combining these with the Magnetohydrodynamic (MHD) convection definition, we will set convection over as wide a range as is possible and conduct recalescence studies from the undercooled melt using the Material Science Laboratory Electromagnetic Levitator (MSL-EML) to compare to results from ground-based EML and ESL. The critical parameters of interest are evaluation of the incubation time between formation of metastable phases and subsequent transformation to the stable phase for Fe-based and titanium aluminide alloys. Since this transformation is very rapid, the high-speed video system must be used to image the processes associated with nucleation and growth during phase selection. 2. Thermophysical Properties. Task description: The density, viscosity, and surface tension of the flight samples will be measured in flight for a range of temperatures. Density will be determined by the video metrology, and surface tension and viscosity by the oscillating drop method. This WP includes determination of parameter modifications required for these measurements as well as collection, reduction, and analysis of the resulting data. The resulting data will support the whole range of theoretical and modeling efforts in this proposal. 3. Modeling of Macroconvection. Task description: Magnetohydrodynamic (MHD) model of macroconvection in the samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts. Collaboration on development of parameters for other work packages to enable the use of convective state as an experimental parameter. Main project: Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order (ICOPROSOL); Science Coordinator is Dr. Dirk Holland-Moritz, Deutsches Zentrum für Luft- und Raumfahrt (DLR), Institut für Materialphysik im Weltraum, Linder Höhe, Köln, Germany. Subproject: Modeling of macroconvection: Magnetohydrodynamic modelling of macroconvection in Ti-Zr-Ni samples for different combinations of experimental parameters. Evaluation of fluid velocity, shear rates, turbulence intensity, and their distribution for control of experiments and input to other theoretical and modeling efforts. Main project: Thermophysical Properties of Liquid Metallic Alloys - Modelling of Industrial Solidification Processes and Development of Advanced Products (THERMOLAB - ISS); Science Coordinator is Professor Hans-Joerg Fecht, Ulm University, Ulm, Germany. Subprojects: Long Duration Microgravity Experiments - ISS. Evaluation of microgravity experiments for fluid flow investigations. MODELLING A: Thermodynamics, Phase Selection. Modelling of nucleation under various flow conditions for steel alloys. MODELLING B: Non-contact thermophysical. Modeling of magnetohydrodynamic convection and shear in 1-g, microgravity for 4-6 alloys. Modeling of Marangoni convection in 1-g ESL drops for comparison with microgravity experiments. Modeling of temperature distribution in 1-g and microgravity samples, 4-6 alloys. Input into design of experimental parameters for 4-6 Alloys.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project consists of modeling in support of three different flight experiments: - PARSEC - Peritectic Alloy Rapid Solidification with EM Convection - ThermoLAB - Thermophysical Properties of Liquid Metallic Alloys - ICOPROSOL - Thermophysical properties and solidification behaviour of undercooled Ti-Zr-Ni liquids showing an icosahedral short-range order In PARSEC, we are investigating the effect of fluid flow on the solidification path of certain alloys. Control of the solidification path would enable tailoring of the microstructure and properties of metal parts for specific applications. In ThermoLAB, we are investigating the thermophysical properties of high-temperature materials. Many of these materials are used commercially. A better understanding of the physical properties will allow more efficient and more reliable production of metallic parts using these alloys. Other alloys are chosen because of scientific interest in their properties, which may illuminate fundamental physics related to the formation of bulk metallic glasses and quasicrystals. In ICOPROSOL, we are investigating the nucleation and growth of quasicrystals, and the effect of atomic-scale order on the macroscopic properties of these alloys. This fundamental investigation may improve our ability to tailor the microstructure of metals for commercial applications.
	Dr. Gwendolyn Bracker joined Worcester Polytechnic Institute (WPI) in October 2024 as an Assistant Research Professor. She continues to collaborate on this project, now under support from the subject grant. • Directed Electromagnetic Levitator (EML) flight operations 6/24 from DLR Cologne, in collaboration with G. Bracker and S. Schneider of the DLR. • The Unified Support for THERMOLAB - ISS, ICOPROSOL, and PARSEC (USTIP) team supported the Investigators' Working Group meetings (IWG) in May 2024 at the European Space Agency (ESA) European Space Research and Technology Centre (ESTEC), and plans to support the IWG meeting in June 2025. • Planned upcoming experiments for ISS-EML Batch 3.5 on Zr-2.5Nb-0.11O. • Planning activities for high-entropy alloys (HEA) measurements on ISS-EML Batch 4 Samples 133 Ni2.1FeCoAlCr and 134 Ni2Fe2AlCr. • Planning activities for upcoming experiments on Sample 135 PdSi in Batch 4. • Planning activities for Ti sample in ISS-EML Batch 5 • G.P.

Task Progress:	Bracker did a theory review on the mathematical model of dynamic nucleation. • Analyzed experimental results of Batch 3.4 dynamic nucleation experiments in GAS and calculated MHD flow. G.P. Bracker supported ICOPROSOL-analyzed growth velocity experiments for an ISS-EML Batch 2 Sample. • G.P. Bracker supported ICORPOSL-analyzed results of ground based ESL measurements on thermophysical properties of compositions used in Batch 2, 3, and (planned) 4. • G.P. Bracker analyzed prior dynamic nucleation experiments for accessibility of growth velocity data. • G.P. Bracker, R. Bradshaw, and Y. Muniz-Diaz explored extending existing machine vision tools for density measurements to growth velocity measurements in preparation for upcoming experiments in ISS-EML and in support of investigations with ICOPROSOL. • G.P. Bracker applied and was awarded cycles for experiments in upcoming TEMPUS campaign. [Ed. Note: The electromagnetic levitation facility, TEMPUS, developed by Dornier, is designed to operate under microgravity conditions.] • G.P. Bracker gathered and is preparing data on Batch 2 experiments for upload to the NASA Physical Sciences Informatics (PSI) Database. • Profs. Bracker and Hyers are both co-organizers of the symposium “Materials Research in Reduced Gravity”, to be held at The Minerals, Metals and Materials Society (TMS) Annual Meeting in March 2026. • Prof. Hyers is a co-organizer of the symposium “Z02: Electrochemical / Materials Processing in Space Engineering 2”.
Bibliography Type:	Description: (Last Updated: 03/30/2026)
Articles in Other Journals or Periodicals	G.P. Bracker, S. Schneider, S. Nell, M. Beckers, M. Mohr, and R.W. Hyers. "Dynamic nucleation in Zirconium-Niobium2.5 during microgravity electromagnetic levitation experiments." Under review as of March 2026. , Mar-2026
Articles in Peer-reviewed Journals	Gangopadhyay AK, Sheng Y, Bracker GP, Hyers RW, Kelton KF. "Slower nucleation kinetics with stirring in a supercooled Zr80Pt20 liquid—Terrestrial and microgravity experiments on the International Space Station." The Journal of Chemical Physics. 2025 Apr 7;162(13). https://doi.org/10.1063/5.0263986 , Apr-2025
Articles in Peer-reviewed Journals	Bracker GP, Nell S, Beckers M, Schneider S, Tonani-Penha LE, Hyers RW. "Suppression of dynamic nucleation by an inert atmosphere in Zr-2.5 Nb-0.11 O in ISS-EML." IOP Conf. Ser.: Mater. Sci. Eng. 2025 Sep 1;1335(1):012002. IOP Publishing. http://dx.doi.org/10.1088/1757-899X/1335/1/012002 , Sep-2025